

Section 10 Solution

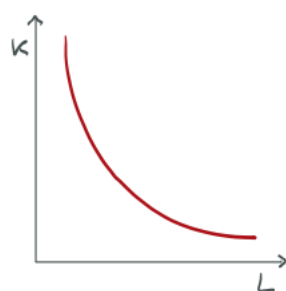
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2 Exercises

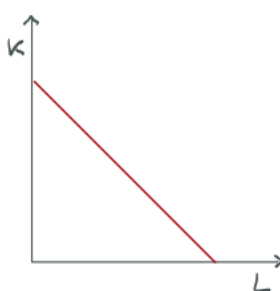
2.1 Isoquants

- Draw one isoquant for each of the following production functions:

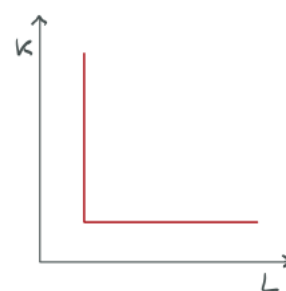
- $y = K^\alpha L^{1-\alpha}$
- $y = K + L$
- $y = \min\{K, L\}$



$$y = K^\alpha L^{1-\alpha}$$



$$y = K + L$$

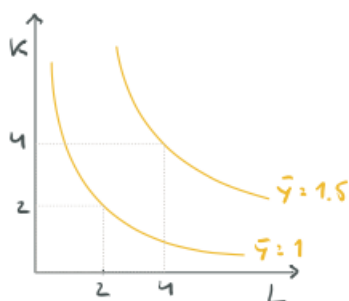


$$y = \min\{K, L\}$$

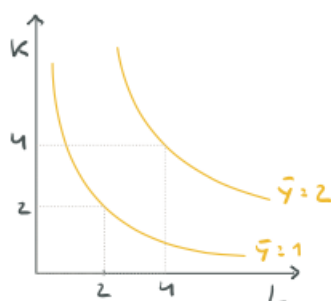
Answer:

- In a new graph, draw the isoquants of two production processes that exhibits (a) constant returns to scale, and (b) increasing returns to scale.

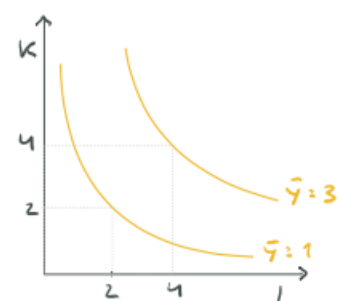
Answer:



DRS



CRS



IRS

*These notes are a consolidation of those of Sam Leone, Pablo Munoz, Yassine Sbair Sassi, Jon Schellenberg, Nick Li, Katalin Springel, Jonas Tungodden, Dan Acland, Justin Gallagher, Mariana Carrera, Matt Leister, Ivan Balbuzanov, Anne Karing, and David Wu, Sam Wang, and Matteo Saccarola.

2.2 Returns to Scale

Determine whether the following production processes exhibit decreasing, constant, or increasing returns to scale.

- Special Case Perfect Substitutes: $y = (L + K)^{1/2}$
Answer: $f(tL, tK) = (tL + tK)^{1/2} = t^{1/2}(L + K)^{1/2} < t(L + K)^{1/2}$ Scaling the original inputs by $t > 1$ generates less production than scaling the original output by t : this production function exhibit *decreasing returns to scale*.
- General Cobb-Douglas: $f(L, K) = L^\alpha K^\beta$
Answer:

$$f(tL, tK) = (tL)^\alpha (tK)^\beta = \begin{cases} t^{\alpha+\beta} L^\alpha K^\beta < tL^\alpha K^\beta & \text{if } \alpha + \beta < 1 \rightarrow \text{Decreasing returns to scale} \\ t^{\alpha+\beta} L^\alpha K^\beta = tL^\alpha K^\beta & \text{if } \alpha + \beta = 1 \rightarrow \text{Constant returns to scale} \\ t^{\alpha+\beta} L^\alpha K^\beta > tL^\alpha K^\beta & \text{if } \alpha + \beta > 1 \rightarrow \text{Increasing returns to scale} \end{cases}$$

2.3 Returns to Scale (Agriculture)

Let $f(L, K, F) = AL^\alpha K^\beta F^\gamma$ where F is farmland.

1. Show that $f(L, K, F)$ exhibits constant returns to scale if $\alpha + \beta + \gamma = 1$. Interpret this result.

Answer: $f(tL, tK, tF) = A(tL)^\alpha (tK)^\beta (tF)^\gamma = t^{\alpha+\beta+\gamma} AL^\alpha K^\beta F^\gamma = t^{\alpha+\beta+\gamma} f(L, K, F)$. This is less than $tf(L, K, F)$ if $\alpha + \beta + \gamma < 1$, greater than $tf(L, K, F)$ if $\alpha + \beta + \gamma > 1$, and equal to $tf(L, K, F)$ if $\alpha + \beta + \gamma = 1$, giving the DRS, IRS, and CRS cases, respectively. This is just a three-variable Cobb-Douglas production function.

2. Consider a small farm with the production function written above, with $\alpha + \beta + \gamma = 1$. In the short-run, however, the farm owner owns a fixed amount of land $F = \bar{F}$. Thus, his short-run production function is $Y_{SR} = AL^\alpha K^\beta \bar{F}^\gamma$. Does this production function have increasing, decreasing, or constant returns to scale in L and K ?

Answer:

Now, F is fixed at $\hat{F}$, giving decreasing returns to scale: $f(tL, tK, \hat{F}) = A(tL)^\alpha (tK)^\beta (\hat{F})^\gamma = t^{\alpha+\beta} AL^\alpha K^\beta \hat{F}^\gamma = t^{\alpha+\beta} f(L, K, \hat{F})$, which is less than $tf(L, K, \hat{F})$ when $\gamma > 0$.